

Footprinting, Reconnaissance and Network Scanning

Ethical Hacking Lab 1

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Portfolio Case Study - Detailed Project Oversight and Completion Record

All offensive-security activities described in this document were performed in authorized, isolated academic laboratory environments.

1. Project Overview

A reconnaissance project combining passive open-source intelligence with authorized active scanning. The work documented domain ownership, hosting and ASN information, geolocation, privacy limitations, and differences between TCP connect, SYN, UDP, and OS fingerprinting scans.

Project Field	Details
Role and Responsibility	Reconnaissance analyst, lab operator, and technical report author.
Environment and Tools	Kali Linux, Nmap, ping, WHOIS/ICANN, ipinfo, ARIN/RIPE/IANA data, authorized Windows and lab targets.
Completion Status	Completed - documented, tested, and included in the master portfolio.
Portfolio Classification	Academic / personal technical project with recruiter-facing case-study documentation.

2. Project Objectives and Success Criteria

- Collect publicly available organizational and infrastructure information.
- Resolve domains, identify hosting providers, and cross-check RIR allocations.
- Compare active scan types and their visibility, speed, and completeness.
- Document ethical boundaries between passive and active reconnaissance.

3. Project Oversight and Execution Process

3.1 Planning and Scope

Defined the authorized environment, project goal, tools, expected outputs, and safety limitations.

3.2 Implementation

Executed the technical tasks and maintained notes, commands, screenshots, and configuration evidence.

3.3 Validation and Testing

Compared actual output with objectives, confirmed repeatable behavior, and collected proof of completion.

3.4 Analysis and Reporting

Converted technical evidence into findings, defensive interpretation, limitations, and recommendations.

3.5 Portfolio Finalization

Prepared a recruiter-friendly summary, detailed case study, screenshots, source report, and public-release guidance.

4. Technical Methodology

1. Resolved selected domains and IP addresses and compared publicly available registration data.
2. Cross-checked geolocation, ASN, hosting, and RIR allocation records.
3. Analyzed privacy redaction and CDN effects on reconnaissance.
4. Ran authorized Nmap TCP connect, SYN, UDP, and OS-detection scans.
5. Compared results and documented when each technique is appropriate.

5. Deliverables and Completion Evidence

Deliverable	Status	Evidence
Technical implementation / lab execution	Complete	Commands, configurations, code, or forensic workflow documented.
Testing and validation	Complete	Results and screenshots included below and in the original report.
Detailed project documentation	Complete	This case-study document and README.
Source/original evidence	Complete	Original report or source files stored in this project folder.
Recruiter-ready description	Complete	Summary and skills section included.
Public publishing review	Required before posting	Remove credentials, personal identifiers, confidential case data, and unnecessary target details.

6. Results and Outcomes

- The project produced a structured passive-intelligence table and an active-scanning comparison.
- It showed that public records can reveal infrastructure relationships while privacy services and CDNs reduce direct attribution.
- Active scan types provided different service visibility and detection trade-offs.

7. Challenges, Decisions, and Lessons Learned

- Avoiding overreliance on geolocation and WHOIS fields that may be stale or privacy-protected.
- Distinguishing a CDN edge address from an origin system.

8. Screenshots and Supporting Evidence

Domain / IP	IP Address	Location	Registrant / Organization	Address & Phone
		Piscataway, New Jersey	University of New Jersey — Office of Information Technology. Registry: EDUCAUSE (.EDU authoritative registry).	Piscataway, NJ 08854-8096, USA. Phone: +1-848-445-4636 (Registrar's Office, New Brunswick campus). Abuse: abuse@rutgers.edu
secretcorp.org	185.199.108.153 185.199.109.153 185.199.110.153 185.199.111.153	United States (hosted on GitHub Pages — Fastly)	Fictitious organisation created by Omar Santos for OSINT training (used in his books, video courses and the Red Team Village). Registrant details redacted via privacy service.	Registrant address & phone: REDACTED (privacy service). Hosting: GitHub Pages / Fastly CDN.

Step 2 – Resolving IPs via ping

Commands used to resolve each target:

```
$ ping -c 1 h4cker.org
PING h4cker.org (185.199.108.153) 56(84) bytes of data.

$ ping -c 1 examcram.com
PING examcram.com (168.146.67.181) 56(84) bytes of data.

$ ping -c 1 rutgers.edu
PING rutgers.edu (128.6.82.29) 56(84) bytes of data.

$ ping -c 1 secretcorp.org
PING secretcorp.org (185.199.108.153) 56(84) bytes of data.
```

Step 3 – WHOIS lookups (example output)

A representative WHOIS lookup for rutgers.edu using whois.net / lookup.icann.org returned:

```
Domain Name: RUTGERS.EDU
Registrant:
  Rutgers, The State University of New Jersey
  Office of Information Technology
  96 Davidson Road
  Piscataway, NJ 08854-8096  USA
Administrative Contact:
  Domain Admin, Office of Information Technology
  Email: hostmaster@rutgers.edu
Name Servers: dns1.rutgers.edu, dns2.rutgers.edu, dns3.rutgers.edu
Registry:  EDUCAUSE (authoritative for .EDU)
```

Step 4 – Geolocation (ipinfo.io)

```
138/udp open|filtered netbios-dgm
500/udp open|filtered isakmp
5353/udp open mdns
```

Interpretation: UDP scanning is slow because most UDP services do not respond to empty packets; nmap has to wait for ICMP 'port unreachable' replies. Many results show open|filtered because the absence of a reply is ambiguous.

Step 9 – OS Fingerprint Scan

```
$ sudo nmap -O 192.168.56.101
```

```
Device type: general purpose
Running: Microsoft Windows 10
OS CPE: cpe:/o:microsoft:windows_10
OS details: Microsoft Windows 10 1909 - 22H2
Network Distance: 1 hop
```

Step 10 – Analysis

- Nmap successfully identified the target as Microsoft Windows 10 with high confidence, based on TCP/IP stack behaviour (window size, TTL 128, options ordering, etc.).
- The port set returned by -sT and -sS was identical, as expected, because both types detect the same listening TCP services; they differ only in how the probe packet is crafted.
- UDP results were less reliable — some ports were reported as open|filtered because no response was received. This is normal behaviour for UDP scans.
- OS fingerprinting returned the correct operating system family and version range. Accuracy would drop if the target had a host-based firewall blocking ICMP or if fewer open and closed ports were available for fingerprinting.
- As a defender, the noisy full-connect and OS scans would leave clear artefacts in Windows Event Log and in an IDS such as Snort/Suricata (SYN floods, half-open connections, OS probe signatures).

Exercise 3 — Active Reconnaissance using the Google Hacking Database

Each dork below is a live query on google.com. Ethical note: queries were crafted only to demonstrate the technique for the lab. None of the retrieved resources were downloaded, credentials were not used, and no further action was taken against the returned sites.

1. Excel file on the internet that contains usernames and passwords

```
filetype:xls OR filetype:xlsx intext:"username" intext:"password"
```

Refined alternative that is commonly shown in the GHDB:

```
filetype:xls inurl:"email.xls"
```

Active scanning and Nmap scan-method comparison.

9. Skills Demonstrated

OSINT, WHOIS, RIR analysis, ASN and hosting analysis, Nmap, TCP/UDP scanning, OS fingerprinting, ethical scoping.

10. Recommendations and Future Improvements

- Add DNS history, certificate transparency, and passive DNS sources.
- Create a repeatable OSINT checklist and evidence log.

11. Portfolio and Resume Presentation

Footprinting, Reconnaissance and Network Scanning - A reconnaissance project combining passive open-source intelligence with authorized active scanning. The work documented domain ownership, hosting and ASN information, geolocation, privacy limitations, and differences between TCP connect, SYN, UDP, and OS fingerprinting scans.

Recommended repository contents: README.md, Documentation/, Evidence/, Source/ or Original_Report/, and a clear authorization/privacy notice.

12. Completion Statement

This project is considered complete for portfolio use because the scope, implementation process, testing evidence, results, skills, limitations, and future improvements are documented. Before public release, sensitive data must be reviewed and removed.